SOURAV DATTO

Ph.D. Applicant | AI/ML for Quantum Networking and Network Optimization

Khulna, Bangladesh | duttosourav8@gmail.com | Google Scholar | GitHub | Portfolio | ORCID

11 peer-reviewed publications | 37 citations | h-index: 5

EDUCATION

American International University-Bangladesh (AIUB) **Nov 2021 - Feb 2026**
B.Sc. in Computer Science and Engineering, Major in Information Systems

- **CGPA: 3.65/4.00.** Relevant coursework: Algorithms, Computer Networks, Machine Learning, Data Science, Linear Programming, Network Security, Wireless Sensor Networks, Calculus I-II.

RESEARCH PUBLICATIONS

- **Reinforcement Learning for Smart Grid Stability Using Adaptive Control and State Abstraction.** *Published at IEEE TENCON 2025.* ↗
  Sourav Datto, Mustakim Ahmed, Md. Eaoumoon Haque, Kazi Redwan, Sajedul Islam, Nasif Hannan, Mohammad Shah Paran, Abu Shufian
- **Enhancing Crop Recommendation with Conformal Prediction and Calibrated Confidence.** *Published at IEEE ICCIT 2025.* ↗
  Sourav Datto, Rafiul Hasan Efti, Md. Abdullah Al Mamun Saykat, Mohaimen-Bin Noor
- **Dynamic and Transformative Hybrid Machine Learning Strategies for Effective Stability Prediction in Smart Energy Networks.** *Published at IEEE ICCIT 2024.* ↗
  Kazi Redwan, **Sourav Datto,** Mustakim Ahmed, Nasif Hannan, Abu Shufian, Mohammad Sakib Mahmood
- **A Multimodal Deep Learning Framework for Integrating Visual, Textual and Categorical Features in Retail Price Estimation.** *Published in Elsevier's Array, Vol. 28, Article 100565 (2025).* ↗
  Kazi Redwan, **Sourav Datto,** Mustakim Ahmed, Habibur Rahman Masum, Md. Faruk Abdullah Al Sohan, Abu Shufian
- **Melanoma Detection Using Augmented ResNet34 for High-Precision Dermoscopic Image Classification.** *Published in Springer CCIS, ICCIES 2025, Vol. 2584, pp. 101-115.* ↗
  Mustakim Ahmed, Arifa Sultana, Prithanjoly Biswas Pew, **Sourav Datto,** Tanzim Ikram Sheikh, Kazi Redwan, Md. Faruk Abdullah Al Sohan

RESEARCH EXPERIENCE

Research Centre for Green Energy, Electro-Anatomy, and Digital Intelligence - Bangladesh **Oct 2024 - Present**
Research Assistant

- Conduct research on smart-grid stability, uncertainty-aware forecasting, and healthcare intelligence, contributing to peer-reviewed studies and reproducible experiments.
- Design preprocessing, statistical modeling, evaluation, visualization, and manuscript-development pipelines across machine-learning and reinforcement-learning projects.

Tech Wing Lab - Bangladesh **Jun 2024 - Present**
Lead Researcher

- Lead collaborative research in machine learning, NLP, computer vision, cybersecurity, and quantum machine learning.
- Mentor junior researchers in experimental design, validation, technical writing, and conference-submission workflows.

ACHIEVEMENTS

- **IEEE TENCON 2025:** Two first-author papers on reinforcement learning and explainable AI.
- **Data Innovators Challenge, MetaData 2024:** Finalist.
- **Robo Carnival 2024, BUET Robotics Society:** Project Showcase participant.
- **IEEE SPAVe 7.0 (2024) and Game of Datathon, Bitfest 2025:** Participant.

PROJECTS

Physics-Informed Machine Learning for Quantum-Network Routing ↗ **2026**
SeQUeNCe, scikit-learn, SHAP, LIME
- Built an 8-router quantum-network simulation with 11 physics-informed features and leakage-safe nested evaluation on a structurally unseen topology.
- Achieved **0.950 external ROC-AUC** and **0.712 MCC**; reduced mean routing regret by **46.5%** and explained decisions with SHAP and LIME.

Quantum Kernel Learning for Multiclass Pattern Recognition **2025**
Qiskit, QSVC, PauliFeatureMap, quantum-noise simulation
- Implemented quantum-kernel classification and Kernel PCA, reaching **93.33% accuracy** and benchmarking against classical SVM baselines.

Conformal Prediction for Calibrated Decision Support **2025**
Distribution-free uncertainty quantification
- Developed calibrated prediction sets for reliable decisions under uncertainty and positioned the method for fidelity-aware quantum-channel estimation.

SecOps Log Anomaly Triage for Distributed Systems **2025**
Isolation Forest, rare-transition analysis, explainability
- Built an HDFS session-level anomaly pipeline with hybrid risk scoring and analyst-facing explanations.

CONFERENCE PRESENTATIONS

- **IEEE TENCON 2025** - Kota Kinabalu, Malaysia; two first-author papers.
- **IEEE ICCIT 2024** - Cox's Bazar, Bangladesh; smart-energy network stability.
- **IEEE ICCIT 2025** - Conformal prediction and calibrated confidence.
- **IEEE QPAIN 2025** - Rangpur, Bangladesh; comparative machine-learning study.
- **ICERIE 2025 and ICDSAIA / ICCIES 2025** - Sylhet, Dhaka, and Ho Chi Minh City.

TECHNICAL SKILLS

Languages: Python, C++, R, SQL
Quantum Networking: SeQUeNCe, entanglement-route simulation, loss/fidelity/memory modeling, route feasibility, policy evaluation
Quantum Computing: Qiskit, QSVC, quantum kernels, PauliFeatureMap, Kernel PCA, quantum-noise simulation
AI/ML: scikit-learn, PyTorch, TensorFlow/Keras, Hugging Face, reinforcement learning, conformal prediction, SHAP, LIME
Data and Research Tools: NumPy, Pandas, Jupyter, Git/GitHub, LaTeX
Human Languages: Bengali (native), English (B2)

RESEARCH INTERESTS

AI-driven quantum networking; entanglement routing and scheduling; fidelity-aware resource allocation; satellite-assisted quantum networks; reinforcement learning; conformal prediction; explainable machine learning.

ACADEMIC REFEREES

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